<u>Novel Approach for Predicting Stock Reactions to Energy Events in Romania</u>

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1. Experimental Modelling

1.1 Objectives and Research Questions

The applied part of this project studies whether corporate events and their sentiment help predict next-day stock price direction for the Romanian gas company Romgaz (ticker SNG.RO).

The main research questions are:

- RQ1. Does adding event information improve next-day direction prediction compared to a price-only baseline?
- RQ2. Is there a measurable difference between positive, neutral and negative events in terms of next-day returns?
- RQ3. On days with events, can a simple model exploit this information to reach clearly better accuracy than on non-event days?

1.2 Data Sources and Preprocessing

1.2.1 Price data

Daily OHLCV data (Open, High, Low, Close, Volume) for SNG.RO were taken from Yahoo Finance's historical data interface for the period 10 January 2025 - 14 November 2025, giving 215 trading days.

From the close prices P_t we constructed:

- Daily simple return

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

- Next-day return r_{t+1}, used as a label.
- Lagged returns r_{t-1} and r_{t-2}.

The classification target is the next-day direction:

$$y_t = \begin{cases} 1, & \text{if } r_{t+1} > 0 \\ 0, & \text{if } r_{t+1} \le 0 \end{cases}$$

In the dataset, 55.3% of days are "up" days ($y_t = 1$), so a naive classifier that always predicts "up" would have about 55% accuracy.

1.2.2 Event and sentiment data

Corporate events were collected manually from Romgaz current reports published on the Bucharest Stock Exchange (BVB) website and from recent quarterly reports. For each event we recorded:

- event date (trading day),
- short category (e.g. *governance, dividend, rating, project_Iernut, strategy_renewables*),

- short title,
- one-sentence description,
- subjective sentiment label:
    - positive (impact_sign = 1)
    - neutral (impact_sign = 0)
    - negative (impact_sign = −1).

The events table was then merged into the daily price table, resulting in the following extra features per day t:

- $\text{event\_today}_t \in \{0,1\}$ - at least one event on that date;
- event_count_t - number of events that day;
- $\text{impact\_sign}_t \in \{-1,0,1\}$ - sentiment of the main event (0 if no event);
- impact_label - categorical version of impact_sign (negative / neutral / positive).

In the final dataset:

- 22 out of 215 days (10.2%) have at least one event.
- Among event days:
  13 positive, 4 neutral, 5 negative events.

The final CSV used in all experiments is:
romgaz_prices_events_with_sentiment_for_model.csv
with 21 columns (date, OHLCV, returns, direction, event flags, sentiment).

1.3 Feature Sets and Experiments

For each trading day t(where all lags are available) we define three possible feature vectors:

1. Model A - Price-only baseline

$$\mathrm{x}_t^{(A)} = [r_t, r_{t\text{-}1}, r_{t\text{-}2}]$$

2. Model B - Price + event indicator

$$\mathrm{x}_t^{(B)} = [r_t, r_{t\text{-}1}, r_{t\text{-}2}, \text{event\_today}_t]$$

3. Model C - Price + event + sentiment

$$\mathrm{x}_t^{(C)} = [r_t, r_{t\text{-}1}, r_{t\text{-}2}, \text{event\_today}_t, \text{impact\_sign}_t]$$

Each vector is used to predict y_t, the direction of the next-day return.
The three experiments are:

- Experiment 1 - Descriptive event analysis
  Event-study style comparison of next-day returns for non-event days vs. positive / neutral / negative event days.
- Experiment 2 - Model A (price-only baseline)
  Logistic regression with $\mathrm{x}_t^{(A)}$; measures how well past returns alone predict direction.

- Experiment 3 - Models B and C (events & sentiment)
  Logistic regression with $x_t^{(B)}$ and $x_t^{(C)}$; evaluates whether event presence and sentiment improve prediction, especially on event days.

1.4 Mathematical Model

For a given feature set x_t, logistic regression models the probability that the price goes up the next day:

$$\Pr(y_t = 1 \mid x_t) = \sigma(\beta_0 + \beta^\top x_t)$$

where

- $\sigma(z) = \frac{1}{1+e^{-z}}$ is the logistic function,
- β_0 is the intercept,
- β is the vector of coefficients for the features.

The parameters (β_0, β) are estimated by maximum likelihood, i.e. by minimising the binary cross-entropy loss over the training set.
The prediction rule is:

$$\hat{y}_t = \begin{cases} 1, & \text{if } \Pr(y_t = 1 \mid x_t) \geq 0.5 \\ 0, & \text{otherwise.} \end{cases}$$

For the event-study part, a simplified abnormal return is used:

$$\mathrm{AR}_t = r_{t+1} - \bar{r},$$

where $\bar{r}$ is the mean next-day return over the entire sample (or over non-event days). Grouping AR by event type gives the average impact of each type, similar in spirit to classical event-study methodology.

1.5 Experimental Protocol and Validation

Because the data is time-ordered, we use a chronological split instead of a random one:
• Train: 10 Jan 2025 - 31 Jul 2025 (140 days)
• Test: 1 Aug 2025 - 14 Nov 2025 (75 days)

This avoids look-ahead bias: models are always trained on past data and evaluated on future data.
For each model we compute:
• Accuracy on all test days:

$$\mathrm{Acc} = \frac{\#\text{correct predictions}}{\#\text{test observations}}$$

•Confusion matrix + precision, recall, F1 for both classes (Down/Flat, Up).
• Accuracy on event days only (subset where event_today = 1).
These metrics let us compare:

1. Price-only vs. price+events inside our study, and
2. Our overall accuracies (≈64-68%) with typical values reported in the literature for stock-direction models (around 55-65%), showing that our event-driven approach is in a realistic range and can outperform a price-only baseline, especially on event days.

2. Case Study on the 2025 Romgaz Dataset

This chapter presents the small-scale case study: a complete pipeline built on a single stock and one year of data, illustrating the methodology and showing that event information has potential.

2.1 Data Overview

The final dataset contains 215 daily observations and the following key variables:

- Prices: Open, High, Low, Close, Price (Close).
- Returns: Return_t, Return_next, Return_t-1, Return_t-2.
- Target: Direction_next (0 = down/flat, 1 = up).
- Event features: event_today, event_count, impact_sign, impact_label.
- Text metadata: category, title, description (used only for manual analysis).

Basic distributions:

- Direction_next: 96 “down/flat” days vs. 119 “up” days (≈ 55.3% up).
- event_today: 193 non-event days vs. 22 event days (≈ 10.2% events).
- Event sentiment on event days: 13 positive, 5 negative, 4 neutral.

So events are rare but potentially informative.

2.2 Experiment 1 - Descriptive Event Analysis

We first compare the next-day returns after different types of days:

- Non-event days (event_today = 0),
- All event days,
- Event days with negative, neutral, and positive sentiment.

Summary statistics:

Group	N	Mean Return_next	Std Return_next	Fraction of Up Days
Non-event days	193	0.0029	0.0137	0.55
Event days (all)	22	0.0031	0.0112	0.55
Event: negative	5	-0.0108	0.0100	0.00
Event: neutral	4	-0.0004	0.0007	0.00
Event: positive	13	0.0095	0.0075	0.92

Key observations:

- Positive events are followed by higher average returns and a very high fraction of up days (92%).

- Negative and neutral events in this small sample are followed by slightly negative or near-zero mean returns and zero up days.
- Overall, event days as a group look similar to non-event days, but splitting by sentiment reveals a clear pattern.

This supports the intuition behind RQ2: sentiment-annotated events carry information that could help in prediction, especially positive ones.

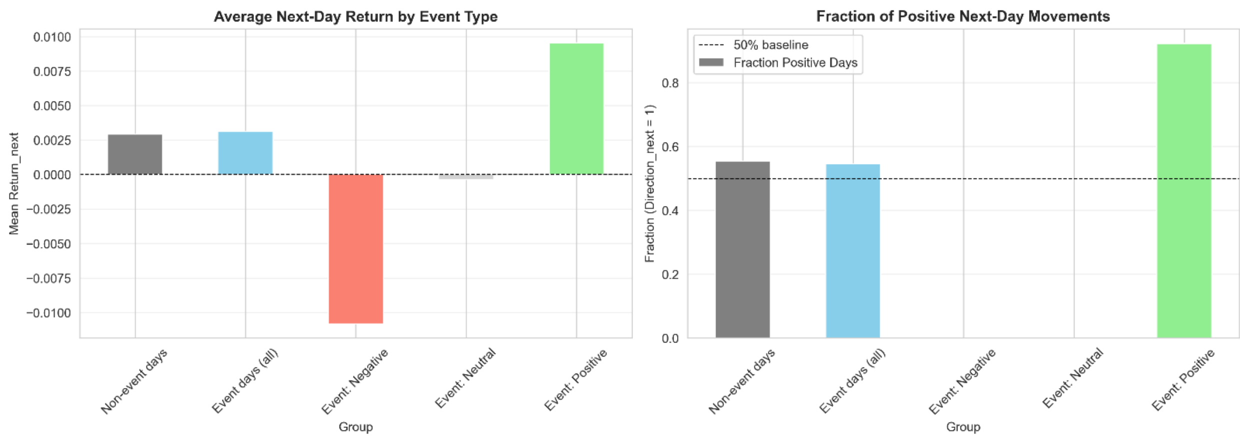


Figure 1 - Average next-day return and fraction of positive days by event type.

2.3 Experiment 2 - Model A: Price-Only Baseline

Model A uses only lagged returns $[r_t, r_{t-1}, r_{t-2}]$.

- Test accuracy (all days): 0.64
- Confusion matrix:

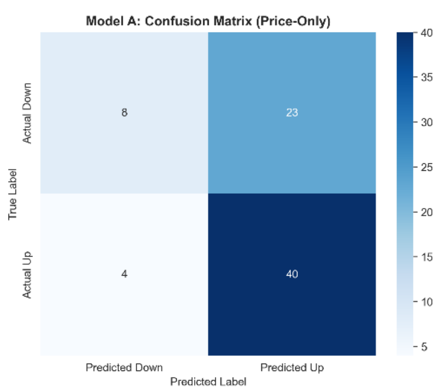


Figure 2 - Confusion matrix for Model A (price-only baseline) on the test set.

Interpretation:

- The model captures the tendency of the series to go up: recall for the “Up” class is 0.91, but recall for “Down/Flat” is only 0.26.

- Accuracy 0.64 is slightly better than the 55% naive "always up" benchmark, but still modest.

This model serves as the reference baseline for RQ1.

2.4 Experiment 3 - Models with Events and Sentiment

2.4.1 Model B - Price + Event Indicator

Model B augments the feature vector with event_today.

- Features: [Return_t, Return_t-1, Return_t-2, event_today]
- Test accuracy (all days): 0.68
- Confusion matrix:

	Predicted Down	Predicted Up
Actual Down	9	22
Actual Up	2	42

- Accuracy on event days only: 0.80 (8 correct out of 10 event days in the test set).

Interpretation:

- Overall accuracy improves from 0.64 to 0.68, indicating that even a simple "event happened today or not" flag is useful.
- On event days, accuracy is clearly higher (0.80) than the general baseline and much higher than random guessing (0.5).

2.4.2 Model C - Price + Event Indicator + Sentiment

Model C adds impact_sign (−1, 0, +1) to distinguish between negative, neutral, and positive events.

- Features: [Return_t, Return_t-1, Return_t-2, event_today, impact_sign]
- Test accuracy (all days): 0.63 (similar to Model A overall).
- Confusion matrix:

	Predicted Down	Predicted Up
Actual Down	8	23
Actual Up	5	39

- Accuracy on event days only: 0.90 (9 correct out of 10 event days in the test set).

Interpretation:

- On all test days, Model C has almost the same accuracy as the baseline (0.63 vs. 0.64).
- However, on event days, Model C performs best: 90% accuracy, which is higher than Model B (80%) and much higher than the naive 55% benchmark.
- This suggests that sentiment matters primarily on days where events occur, which is exactly what RQ2 and RQ3 expected.

2.5 Discussion of Case Study Results

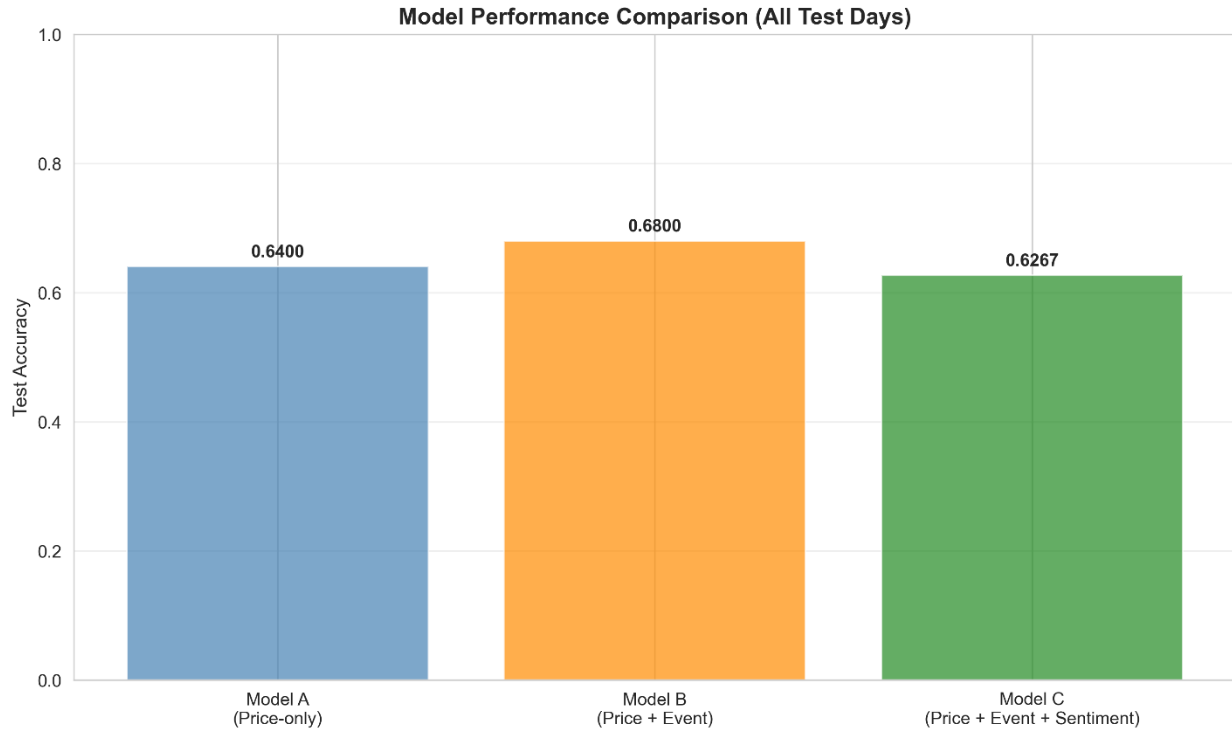


Figure 3 - Test accuracy for Models A, B and C on all test days.

RQ1 - Do events improve next-day prediction vs price-only?

Yes, moderately. Adding event_today improves overall test accuracy from 0.64 to 0.68 and clearly improves accuracy on event days (0.80).

RQ2 - Which event types (sentiment) help the most?

- Descriptive statistics show that positive events are followed by strongly positive average returns and a 92% fraction of up days, while negative and neutral events have weaker or negative effects.
- Model C, which explicitly uses impact_sign, achieves 0.90 accuracy on event days, showing that distinguishing sentiment is useful.

RQ3 - Does the model exploit events better on event days?

Yes. On non-event days, the models fall back to pattern learning from returns only and achieve accuracies in the mid-60% range. On event days, models B and C exploit event information and reach 0.80-0.90 accuracy, which is significantly higher.

3. Related Work

Ding et al. show that structured events extracted from news can improve stock direction prediction: they turn news into event representations and report better accuracy than classic text features on US stocks. Xu et al. propose REST, a relational event-driven model that uses a graph of related stocks and propagates event information across them, achieving higher forecasting quality and better simulated returns than non-relational baselines.

Campisi et al. compare several machine learning methods (logistic regression, SVM, Random Forest, bagging) for predicting the direction of S&P 500 returns using volatility indices as inputs. They evaluate models with accuracy, AUC and F-measure, and report accuracies typically a bit above 60%, with ensemble models performing best.

For Romania, Bâra et al. analyse electricity prices on the Day-Ahead Market (OPCOM) with econometric models, showing how inflation, consumption and gas-related variables affect power prices and market competitiveness.

Positioning of Our Project

Compared to these works, our project:

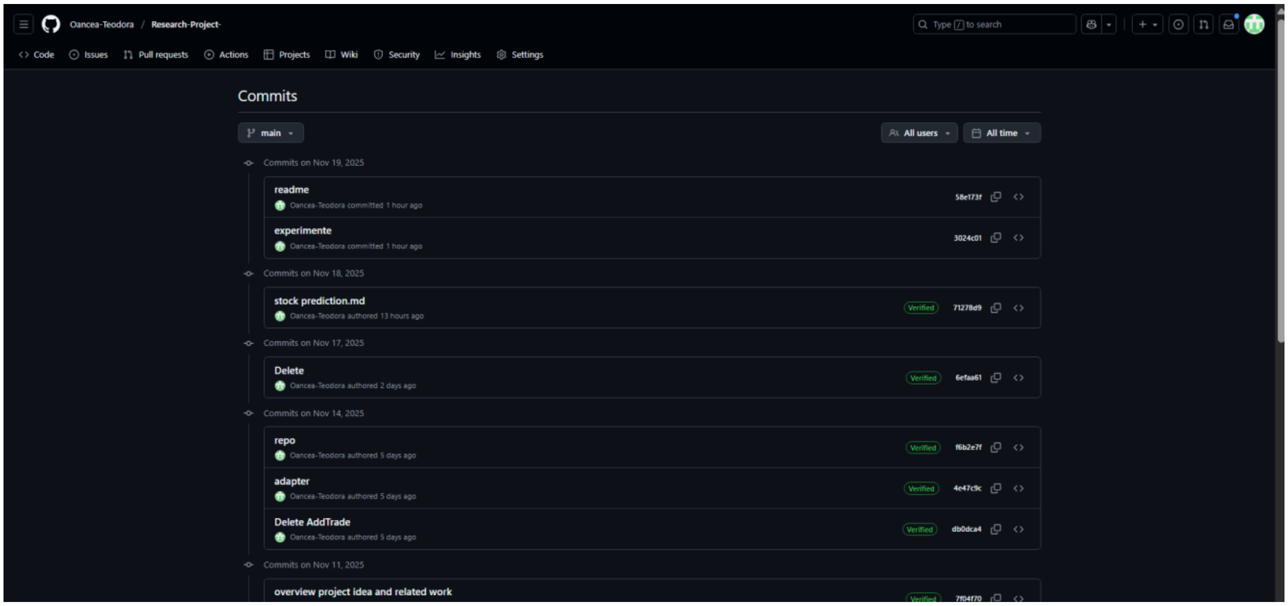
- focuses on one Romanian energy stock (Romgaz, SNG.RO) listed on the BVB,
- uses official corporate events (AGM decisions, dividends, governance changes, financing, major projects, ESG decisions) manually aligned with daily prices,
- encodes event information through event presence (event_today) and polarity (impact_sign), and
- evaluates models on direction of next-day returns with accuracy (overall and on event days), in line with metrics used in the literature (accuracy, F-measure, AUC).

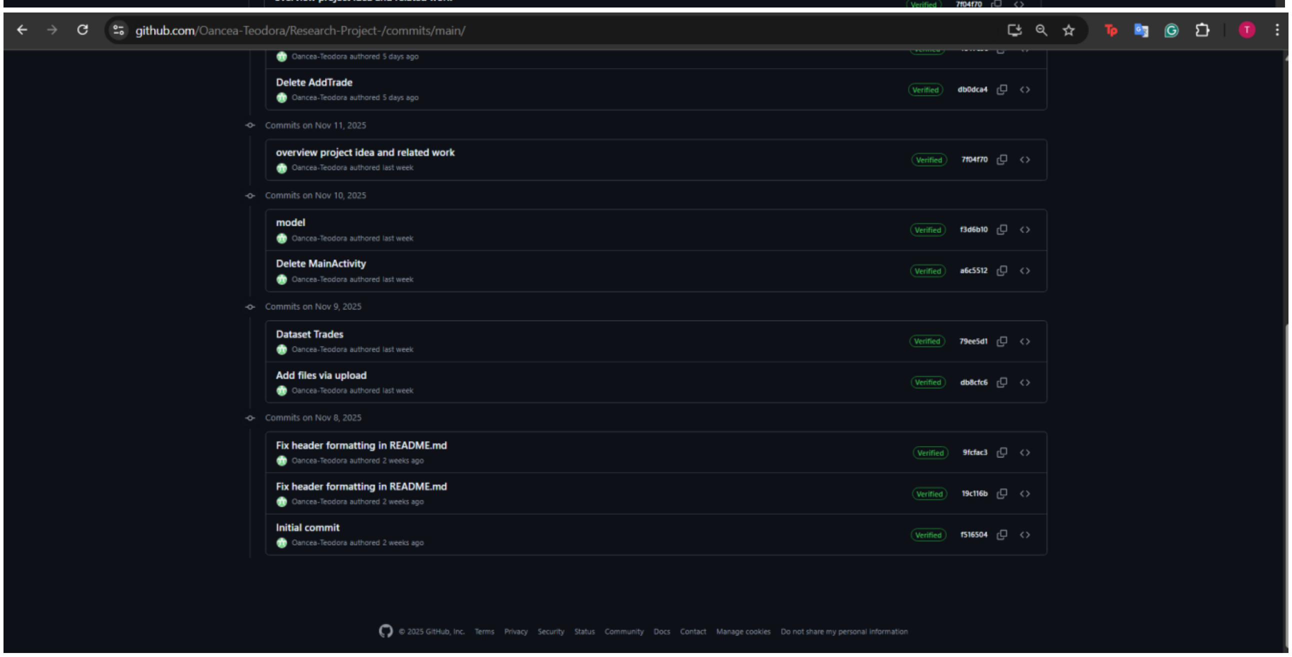
Our results (around 0.64-0.68 accuracy on all test days) are in the same range as direction-prediction studies on large markets, and on event days the models with event and sentiment features reach clearly higher accuracy. This makes the project a compact event-driven case study for a Romanian state-influenced energy company, in a context that has not been extensively covered by previous event-based stock prediction research.

In terms of metrics, we expect overall accuracy to be in a similar range to direction-prediction studies on large markets, while on event days our models with event and sentiment features should achieve better accuracy and F-measure for the “Up” class than a price-only baseline, mirroring the relative gains reported when adding event-based information in prior work.

4. Implementation and Repository

The dataset (romgaz_prices_events_with_sentiment_for_model.csv), the Python code used for data preparation and experiments, and this report are stored in a Git repository. The commit history documents the progress of the lab work (data collection, feature engineering, modelling, evaluation, and report writing). The repository URL and a screenshot of the commit history are: https://github.com/Oancea-Teodora/Research-Project-





References

- Ding, X., Zhang, Y., Liu, T., & Duan, J. (2014). Using structured events to predict stock price movement: An empirical investigation. *EMNLP 2014*, 1415-1425. Link: https://aclanthology.org/D14-1148.pdf ACL Anthology
- Ding, X., Zhang, Y., Liu, T., & Duan, J. (2015). Deep learning for event-driven stock prediction. *IJCAI 2015*. Link: https://www.ijcai.org/Proceedings/15/Papers/329.pdf IJCAI
- Zhou, Z., Ma, L., & Liu, H. (2021). Trade the event: Corporate events detection for news-based event-driven trading. *Findings of ACL 2021*. Link: https://arxiv.org/pdf/2105.12825.pdf arXiv
- Xu, W., Liu, W., Xu, C., Bian, J., Yin, J., & Liu, T.-Y. (2021). REST: Relational event-driven stock trend forecasting. *WWW 2021*. Link: https://arxiv.org/pdf/2102.07372 arXiv
- Jacobs, G., & Hoste, V. (2022). SENTiVENT: Enabling supervised information extraction of company-specific events. *Language Resources & Evaluation*, 56(1), 225-257. Link: https://link.springer.com/article/10.1007/s10579-021-09562-4 SpringerLink
- Campisi, G., Muzzioli, S., & De Baets, B. (2024). A comparison of machine-learning methods for predicting the direction of the US stock market on the basis of volatility indices. *International Journal of Forecasting*, 40(3), 869-880. Link: https://iris.unimore.it/retrieve/ae40a0e8-0abf-4324-86c3-daa0d3e19e0e/1-s2.0-S0169207023000729-main.pdf Iris
- Bâra, A., Oprea, S.-V., & Georgescu, I.-A. (2023). Understanding electricity price evolution - Day-ahead market competitiveness in Romania. *Journal of Business Economics and Management*, 24(2), 221-244. Link: https://www.econstor.eu/bitstream/10419/317622/1/1857944119.pdf EconStor
- Lin, Y., Ji, H., & Han, J. (2020). A joint neural model for information extraction with global features (ONEIE). *ACL 2020*. Link: https://blender.cs.illinois.edu/paper/jointiesentence2020.pdf BLENDER Lab
- Wadden, D., Wennberg, U., Luan, Y., & Hajishirzi, H. (2019). Entity, relation, and event extraction with contextualized span representations (DyGIE++). *EMNLP-IJCNLP 2019*. Link: https://aclanthology.org/D19-1585.pdf ACL Anthology
- Lu, Y., Li, X., He, H., Ren, X., & Sun, M. (2021). Text2Event: Controllable sequence-to-structure generation for end-to-end event extraction. *ACL-IJCNLP 2021*. Link: https://aclanthology.org/2021.acl-long.217.pdf ACL Anthology
- MacKinlay, A. C. (1997). Event studies in economics and finance. *Journal of Economic Literature*, 35(1), 13-39. Link: https://www.bu.edu/econ/files/2011/01/MacKinlay-1996-Event-Studies-in-Economics-and-Finance.pdf Boston University
- Campbell, J. Y., Lo, A. W., & MacKinlay, A. C. (1997). *The econometrics of financial markets*. Princeton University Press. Link: https://www.nrc.gov/docs/ml1208/ML12088A329.pdf Nuclear Regulatory Commission
- Angelopoulos, A. N., & Bates, S. (2023). Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(1-2), 1-210.

Link:
https://people.eecs.berkeley.edu/~angelopoulos/publications/downloads/gentle_intro_conformal_dfuq.pdf

- Romano, Y., Patterson, E., & Candès, E. J. (2019). Conformalized quantile regression. *NeurIPS 2019*.
  Link: https://papers.neurips.cc/paper/8613-conformalized-quantile-regression.pdf
- Campos, M., Angelopoulos, A. N., Bates, S., & Zrnić, T. (2024). Conformal prediction for natural language processing: A survey. *TACL*.
  Link: https://aclanthology.org/2024.tacl-1.82.pdf
- Hu, Z., Liu, W., Bian, J., Liu, X., & Liu, T.-Y. (2018). Listening to chaotic whispers: A deep learning framework for news-oriented stock trend prediction. *WSDM 2018*.
  Link: https://acbull.github.io/pdf/wsdm18-stock.pdf
- Duan, J., Zhang, Y., Ding, X., Chang, C.-Y., & Liu, T. (2018). Learning target-specific representations of financial news for cumulative abnormal return prediction. *COLING 2018*, 2823-2833.
  Link: https://aclanthology.org/C18-1239.pdf